Lab 2: Random Processes
ISC 4221
September 3, 2021

Programming Language: Matlab or Python
Due date: 11:59PM on September 17, 2021
Submit online through Canvas
To receive full credit, you must submit:

1. Well-commented and working code
2. A single PDF writeup containing any plots, discussion, calculations, and conclusions.

Goals: The objectives of this lab are to analyze a random number generator and to use a random process to simulate a game show.

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1. Random Number Generator (40 points)

(i) Generate 10,000 uniform random coordinates, $x_i \in X = [0, 1]$ and $y_i \in Y = [0, 1]$, store them in an array, and plot them.

(ii) Generate histograms of the random samples and show them in the same plot. Your results should resemble Figure 1.

(iii) To determine the randomness of the samples, we can perform a test with a summary statistic. Calculate the means, $\overline{x}$ and $\overline{y}$, of the two samples. Compare your results to the expected value of the two random variables which are both 0.5.

(iv) Create a function that takes in an array of random points with size $N \times 2$ and returns:

- the mean of each of the columns of the array
- the root-mean-squared-error defined as

$$\text{RMSE} = \sqrt{(\bar{x} - 0.5)^2 + (\bar{y} - 0.5)^2}.$$

(v) Use your code to compute the RMSE for N uniformly random coordinates, with $N = 10, 11, 12, \ldots, 10,000$. For each N, you only need to consider one random array of size $N \times 2$, but you must create a new random array for each value of N. Plot the RMSE as a function of N on a loglog scale. If you have done it correctly, you should observe that the RMSE is proportional to $N^{-1/2}$. This is a direct consequence of the Central Limit Theorem. Superimpose a plot of this line on the same loglog scale. Use whatever constant of proportionality is necessary so that the two plots are close.

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2. Monty Hall (60 points) Suppose you are on a game show where there are three doors. Behind one of the doors is a new car, and behind the other two are goats. The host gives you a chance to win the car by telling you to pick a door. Assume you pick door number 1. Before revealing if you have won, the host, who knows which door the car is behind, opens one of the other doors to reveal a goat. They then give you the opportunity to either keep

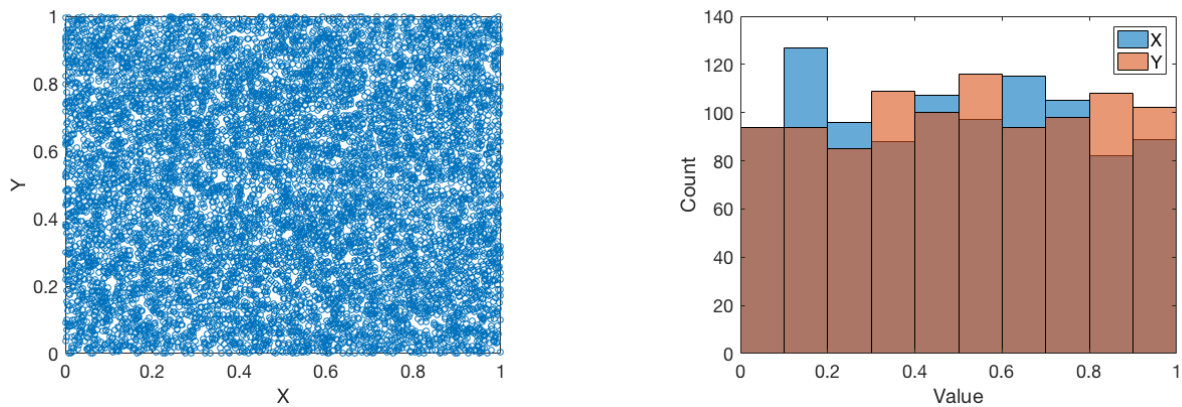

Figure 1: An example of the expected plots

your original door or to switch to the other closed door. Here, we would like to address whether it is advantageous to switch or not, or if it does not matter.

(i) Assume that you pick door 1. Since there are three possible locations of the car and two choices (switch or keep), there are a total of six possible outcomes. For example, the car might be behind door 2, this forces the host to reveal the goat behind door 3, and you decide to keep your original door. In the case that both remaining doors contain a goat, the host chooses one at random to reveal. List all six outcomes and determine the result (do you win a goat or a car) for each of them. What is the probability of winning the car for each of the two strategies?

(ii) Write a program that tests the two strategies. Randomly choose an integer 1, 2, or 3 that represents the door that the host puts the car behind. Again, choose a random integer 1, 2, or 3 to represent the door the contestant chooses. Then, the host picks one of the remaining two doors that contains a goat (this is done randomly if necessary). Finally, record if the car is behind the original door (keeping strategy) or the other closed door (switching strategy). Run your code 1,000 times. How many times did the switching strategy result in a win? How many times did the keeping strategy result in a win? Do your analytic and simulated results agree?